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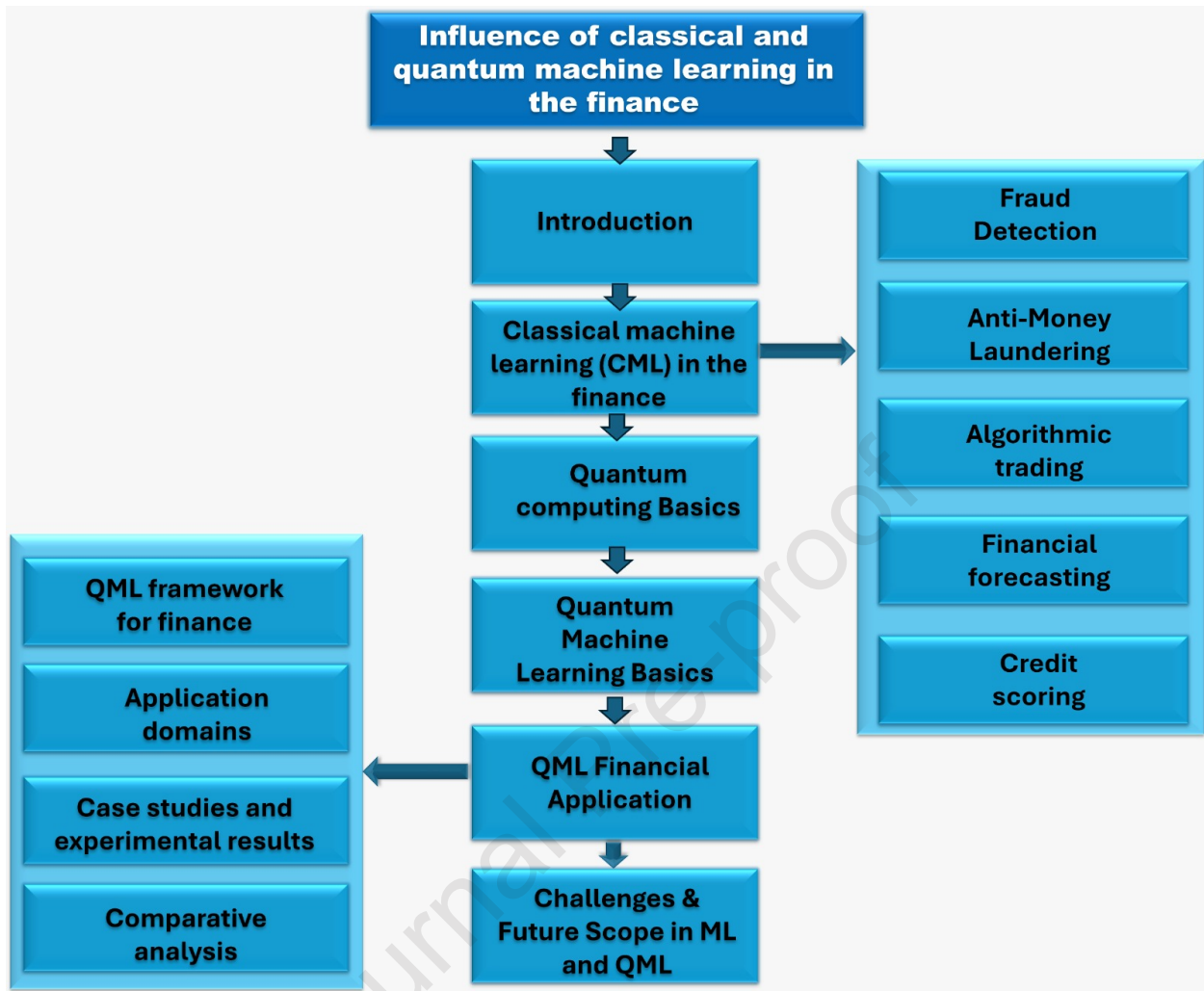
Systematic Review on the Influence of Classical and Quantum Machine Learning in the Financial Sector

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Abstract

In the financial sector, machine learning (ML) has emerged as a transformative technology that provides sophisticated solutions to critical challenges, such as in fraud detection, anti-money laundering, algorithmic trading, credit scoring, and financial forecasting. ML significantly improves the accuracy, speed, and efficacy of decision-making processes by using sophisticated algorithms and large datasets. Nevertheless, its integration into financial systems presents significant obstacles such as cybersecurity threats, regulatory compliance, and data quality concerns. Building on these developments, this review examines the emergence of quantum machine learning (QML), an approach that integrates quantum-computation principles with intelligent financial systems. QML exploits quantum phenomena, including superposition, entanglement, and parallelism, to overcome the computational constraints encountered in classical ML. Applications such as quantum support vector machines (QSVMs), quantum neural networks (QNNs), and quantum clustering are examined in the context of fraud detection, portfolio optimization, and market forecasting. In contrast to previous reviews, this study provides a comparative and integrative analysis of ML and QML in the financial domain. It highlights the current technical limitations and emergent hybrid quantum classical models, emphasizing strategic directions for future research. In addition, we suggest a new framework for assessing the suitability of QML in practical financial applications, thereby bridging the gap between theoretical potential and real-world deployment. This review not only consolidates recent advancements but also establishes the foundation for the next iteration of intelligent financial systems powered by quantum-enhanced technologies.

Keywords: Machine learning, Quantum machine learning, Financial intelligence, Hybrid quantum-classical model, Explainable quantum AI, Quantum optimization

1. Introduction

In recent years, efforts to combat global crime, particularly financial crime, have significantly increased in accordance with the United Nations' Sustainable Development Goal 16, which is dedicated to the development of robust institutions, justice, and peace (Arthur et al., 2025; Knox and Orazgaliyev, 2024). In response to the ever-changing challenges, the financial sector, which is tasked with the management, creation, and research of money and investments, has consistently been at the vanguard of technological innovation (Allioui and Mourdi, 2023). Machine learning (ML) is among the most revolutionary of these technologies, having revolutionized conventional methods of handling financial intricacy by enabling data-driven insights, automation, and improved scalability in a variety of financial services (Nguyen et al., 2023). ML enables decisions to be made in real time with greater accuracy and efficiency by evaluating and processing large amounts of structured and unstructured data. Financial forecasting, algorithmic trading, credit scoring, fraud detection, and anti-money laundering (AML) are real-world applications (Mallika and Ramasubramanian, 2025). These developments enable financial institutions to forecast trends more precisely, reduce risk, spot irregularities, and decrease operational latency. For instance, ML-based fraud detection systems use advanced classification algorithms and neural networks to identify questionable transaction patterns, which lowers financial losses and promotes confidence in online financial systems (Ali et al., 2022). AML systems use clustering and classification techniques to identify anomalies in financial behavior and improve system transparency and regulatory compliance. By leveraging the speed and accuracy of ML, algorithmic trading platforms allow organizations to perform large numbers of transactions at a high frequency, thereby improving market liquidity and trading results (Challa, 2025).

Building on these developments, quantum machine learning (QML) has recently become a cutting-edge area in financial computation. Based on quantum phenomena such as superposition, entanglement, and parallelism, QML eliminates the computational limitations of classical systems (Khan and Robles-Kelly, 2020) by combining ML concepts with the advantages of quantum computing. Using QML algorithms, high-dimensional financial data can be processed with remarkable speed and accuracy. The significant advancements that show promise in the domains of

financial forecasting, portfolio optimization, credit risk assessment, and fraud detection, include quantum support vector machines (QSVMs), quantum neural networks (QNNs), and quantum clustering (Alomari and Kumar, 2023; Beer et al., 2020; Elsedimy et al., 2024). For instance, QSVM can be used to implement quantum kernel methods and recognize intricate patterns in financial datasets, whereas QNNs demonstrate enhanced predictive accuracy when assessing market volatility and creditworthiness. Despite their immense potential, the implementation of ML and QML in financing faces obstacles, with data quality issues, algorithm interpretability, cybersecurity threats, regulatory compliance, current instability, and limited scalability of quantum hardware being among the most significant challenges (Zhou et al., 2025; Zhou et al., 2017). These constraints emphasize the necessity for ongoing research and technological advancement, particularly in the areas of quantum error correction, intelligent automation in finance, and development of hybrid quantum–classical systems that are both resilient and ethically sound.

The objective of this review is to offer an exhaustive, comparative, and forward-thinking examination of ML and QML applications in the financial sector. Our review distinguishes itself from previous studies that analyzed these technologies in isolation, by assessing their practical viability, integrating their overlapping potential, and devising a strategic roadmap for their integration into next-generation financial systems. This study provides a unique perspective that is pertinent and relevant, aligning with the global agenda for the construction of a resilient, intelligent, and secure financial infrastructure, in synthesizing recent advancements and identifying critical voids in research and implementation.

2. Classical machine learning (CML) in the finance sector

Finance encapsulates the management, creation, and study of money and investments. ML algorithms are used to effectively handle financial problems. These algorithms help improve the financial decisions made by people and provide a way for a better future in the finance sector (Singh et al., 2022). ML algorithms have several applications in finance. ML can handle massive amounts of data and produce better results than established methods, thereby increasing the effectiveness of the process (Ahmed et al., 2023). ML applications in finance include fraud detection, AML, algorithmic trading, financial forecasting, and credit scoring. This section outlines each

application, emphasizing the limitations, research directions, and provide recent advancements.

2.1. Fraud detection

Some people break the law by using credit card data for their purchases. Credit card usage has significantly increased, which has also increased fraud opportunities (Seera et al., 2024). The development of e-banking has resulted in billions of losses annually due to credit card fraud (CCF), which refers to unauthorized transactions using stolen card details (Mosa et al., 2024). In the United States, fraud has a significant impact on the economy, costing billions of dollars annually. Therefore, considering the surge in digital transactions and credit card usage, fraud detection remains a critical concern. According to recent reports, India's fraud-detection system has prevented over 1.35 million fraudulent calls daily, protecting assets worth 2,500 Crore INR (Business Today, 2024). Traditional rule-based fraud-detection systems are becoming increasingly ineffective because of evolving fraud techniques. However, ML-based fraud-detection models offer predictive capabilities for identifying anomalies in real time (Aljunaid et al., 2025). The following discussion explains how ML helps reduce fraud detection.

Fraud detection systems use prediction algorithms to differentiate transaction patterns. Fraud detection aims to stop fraud before it occurs. ML enables detection of fraudulent transactions in real time, alerting banks upon identifying fraud patterns (Alarfaj and Shahzadi, 2024). Classification algorithms, such as support vector machines (SVMs), k-nearest neighbors (KNN), logistic regression, and convolutional neural networks (CNNs), have helped identify or classify transactions based on suspicious patterns. The accuracy of these models is assessed using performance metrics, whereby the best model is selected through comparisons (Ito et al., 2021). CNN is applied to identify unusual customer patterns. Fraud can also be detected using autoencoders, which are trained on non-anomalous data to identify fraud anomalies based on reconstruction errors. Restricted Boltzmann machines can also effectively identify fraudulent transactions in larger datasets.

Other popular algorithms for fraud detection include human-in-the-loop and artificial neural networks (ANNs) (Alarfaj et al., 2022). However, they face challenges

such as evolving fraud techniques, high false positives, data privacy and confidentiality, unstructured data, and model complexity (Razaque et al., 2025). These challenges highlight the multifaceted nature of fraud detection and explain why this field requires further research and progress. Future studies should focus on unstructured data sources and advanced text-mining techniques. The novelty of this domain lies in exploring hybrid deep-learning models, graph-based fraud detection, and federated learning techniques, which offer secure and privacy-preserving solutions. Fig. 1 illustrates the fraud detection process using ML techniques.



Fig. 1. Step-by-step workflow of fraud detection using machine learning, highlighting key stages.

2.2. AML

Money laundering is one of the most significant problems worldwide. Money laundering is a global financial crime that threatens economic stability. The UN established 17 sustainable development goals (SDGs) in 2015 (Halişçelik and Soytaş, 2019), of which AML is a key objective, with an estimated annual value between \$800 billion and \$ 2 trillion. This includes organized and violent crimes under the UN's SDG 16 (Arthur et al., 2025). Money laundering is a major crime that refers to illegal currency processing to access a legal financial system. Money laundering constitutes 2% – 5% of the global gross domestic product (GDP) and poses a significant threat to the world. To prevent money laundering, governments have begun to implement AML techniques to combat crime. The Indian government proposed the Prevention of Money-Laundering Act of 2002 (Narayan, 2019). However, manual detection of suspicious activities is not feasible because of the large transaction volumes. Rule-based systems also fail to detect complex transaction patterns. Therefore, researchers have begun leveraging the power of ML, as it offers better scalability, precision, speed, dynamic adaptation, improved detection accuracy, and a collaborative framework. Traditional AML methods struggle with scalability and adaptability, whereas ML offers

a dynamic framework that uses unsupervised clustering, classification algorithms, and deep neural networks (Ezugwu et al., 2022).

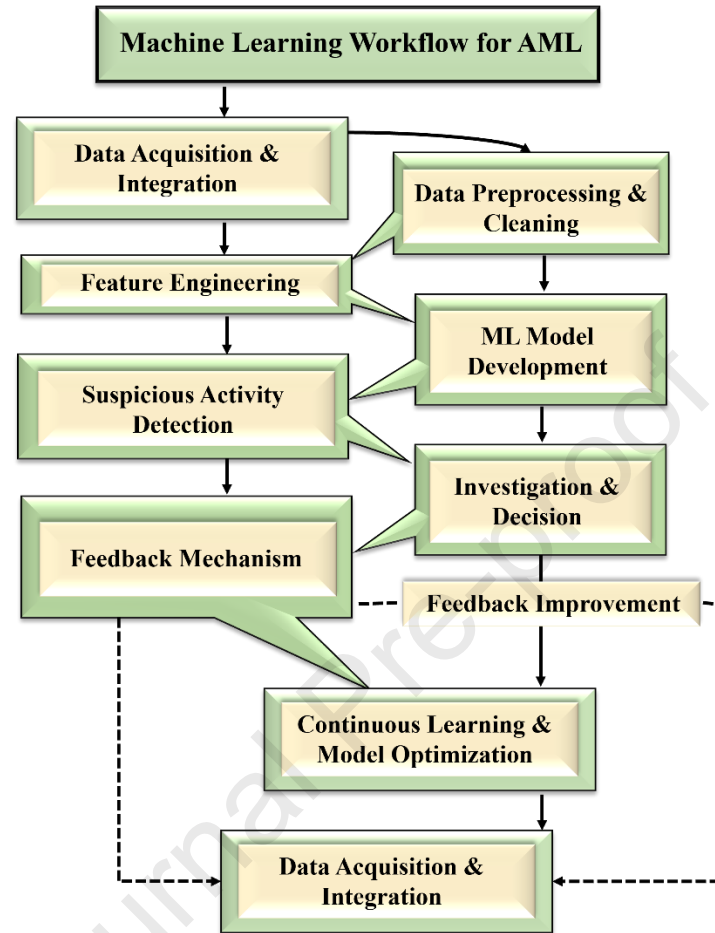


Fig. 2. Machine learning workflow for anti-money laundering, showing steps from data collection to feedback improvement.

In ML, a clustering algorithm is used to group related transactions. By exploring the transaction patterns, clustering can help detect unusual behaviors that can be attributed to money laundering. This method helps categorize transactions based on similarities and makes it easy to identify anomalies (Yang et al., 2023). Classification algorithms are used to estimate money laundering within a given group, allowing prioritized investigation (Mallika and Ramasubramanian, 2025; Zhang and Trubey, 2019). Neural networks can process complex patterns in transaction data, making them suitable for identifying money laundering that may not be detected by simple classical methods. This implementation of the ML technique allows real-time analysis of transactions. Feedback mechanisms refine the ML model over time, improving its

accuracy and effectiveness (Jensen and Iosifidis, 2023). Although this has many advantages, challenges remain, including data quality issues, corrupted or noisy data, the dynamic nature of money laundering tactics, operational costs, Byzantine agreement problems, and regulatory compliance (Khan et al., 2025). To mitigate these challenges, advanced ML techniques must be developed. Future research directions include explainable AI (XAI) models, graph-based AML detection, and integration with natural language processing for unstructured data analysis. Fig. 2 illustrates the ML workflow for AML, outlining the key steps involved in identifying and analyzing suspicious financial activities.

2.3. Algorithmic trading

Algorithmic trading begins with data collection, which involves the acquisition of relevant market data from financial data sources, followed by preprocessing data, where the collected data are cleaned, normalized, and transformed for analysis. The process then moves to defining the trading, in which trading rules and strategies are formulated, and algorithm selection and training, where suitable algorithms are selected and trained using historical data. Finally, execution of trades is carried out by deploying the trained model to automatically place and manage trades in real-time market conditions. Algorithmic trading automates trade execution by using predefined rules and data patterns. The wide range of methods used in algorithmic trading include traditional statistical techniques and more sophisticated ML approaches (Cohen, 2022). While classical strategies such as trend following and mean reversion have acquired traction, recent advances in the field consider ML techniques, particularly deep learning (DL) and reinforcement learning (RL), which enhance decision making by dynamically adapting to market changes (Mosavi et al., 2020). Algorithmic trading is characterized based on:

- Speed and efficiency: Algorithms can process enormous amounts of data and trade much quicker than humans, thus ensuring prompt market response.
- Reduced emotional bias: Automated trading eliminates emotional choices and leads to trading strategies that are more rational.

- **Backtesting capabilities:** Traders can test their strategies for performance using historical data before releasing them for deployment in live markets. Despite all the benefits, algorithmic trading is beset with several problems, such as model generalization, overfitting, and data accessibility (Tudor and Sova, 2024).
- **Market complexity:** Financial markets are remarkably intricate, unforeseeable, and totally unpredictable, making it extremely difficult to prepare an algorithm that works flawlessly every time.
- **Data quality and accessibility:** Algorithmic trading strategies depend on the standards and availability of historical information; the quality and availability of this information vary in different markets.
- **Regulatory scrutiny:** Increased use of algorithmic trading leads to increased scrutiny of trading by the regulatory bodies regarding market transparency and fairness.

Based on the findings of modern technological development and complex algorithms, algorithmic trading is considered one of the most outstanding innovations within the finance industry (Lee, 2020). Despite the advantages of high speed and reduced emotional bias, algorithmic trading poses numerous challenges that require constant research and adaptation. With the evolution of FinTech, algorithmic trading is being featured as a core characteristic of the future finance industry (Mosavi et al., 2020). Novel areas such as meta-learning, multi-agent RL, and sentiment-aware trading bots are gaining traction in academia and industry, signaling a promising future for ML-driven trading.

2.4. Financial forecasting

Financial forecasting involves predicting market trends, asset prices, or economic indicators. In financial forecasting, ML is used to predict loss-related financial analysis and digital crimes affecting financial systems within different contexts considering economic risk-based scenarios (Chang et al., 2024a). QML involves an integrated workflow comprising data acquisition and collaboration, natural language and data preprocessing, experimental or model formulation, quantum circuit-based learning and optimization, and intelligent decision-making through AI-driven analysis. The

constantly growing amount of data generated daily in financial markets has led to employing ML algorithms to process, analyze, and predict market trends, as opposed to traditional statistical techniques. Textual data, along with historical price information and news events, also need to be incorporated into financial forecasting of market dynamics to improve prediction accuracy (Ivanyuk, 2024). This range of strategies in the financial forecasting arena is well represented by the variety of ML methodologies employed, including quantile regression forest modeling, gradient boosting, Bayesian inference, hybrid auto-regressive integrated moving average (ARIMA)-ANN approaches, and synthetic penalized logit boost methods that uncover nonlinear relationships in massive datasets (Duan et al., 2022; Nayak et al., 2022). Over the past few years, ML has become an important tool in financial forecasting, largely because of its ability to analyze intricate datasets and uncover nonlinear relationships. Numerous methodologies have been explored to enhance prediction accuracy within this domain.

Neural networks are used to model nonlinearity in financial data and have proven quite effective in terms of stock price forecasting and mutual fund net asset values, whereas quantile regression forests (QRFs) offer robustness in forecasting under uncertainty (Nayak et al., 2022). Notable applications include dendritic neuron models (DNMs) that use chemical reaction optimization for increased prediction accuracy (Nayak et al., 2022). Even methods such as random forest and gradient boosting that are used for tasks such as growth rate predictions, surpass traditional models in performance. Combined with multiple decision trees or models, these techniques can reduce the possibility of overfitting and boost model stability (Gogas and Papadimitriou, 2021). In the context of financial forecasting, the integration of several ML techniques, including combining ARIMA with neural networks, has been promising (Mochurad and Dereviannyi, 2024). Consequently, the incorporation of advanced ML techniques in financial forecasts not only improves the precision of predictions but also opens opportunities for further research into optimizing these models for better performance. The implementation of ML in financial forecasting enables more accurate systemic risk extraction within financial markets and improves prediction performance (Mashrur et al., 2020). Text mining, news sentiment analysis, and social media analytics have been increasingly integrated to enhance prediction

reliability. The novelty lies in real-time adaptive forecasting models, multimodal data fusion, and probabilistic DL approaches (Das and Singh, 2023).

2.5. Credit scoring

2.5.1. Data sources and processing

Credit scoring assesses a borrower's likelihood of repaying a loan and was one of the earliest applications of ML in the economic field. In contrast to classical credit scoring techniques, which depend heavily on past repayment history, ML-based credit scoring models use a variety of nontraditional datasets (La Gatta et al., 2025). These datasets can include transactions, agricultural productivity, satellite images, local economic data, and in some cases, social network activity to develop credit profiles for people with no credit history. For example, a study by a Chinese FinTech firm established that ML combined with big data significantly boosts the accuracy of credit scoring compared to traditional methods (Abbas, 2025). These data-driven techniques acquire value during times of economic disruption when adaptive real-time borrower risk assessment becomes essential.

2.5.2. ML models for credit scoring

In modeling techniques, ML credit scoring leverages a variety of algorithms to capture complex borrower–credit relationships. Decision trees are widely used to analyze borrower information because of their ability to handle numerical and categorical variables (Wang et al., 2021; Ziemba et al., 2023). Neural networks exhibit better predictive performance because they capture the nonlinear relationships between input variables and credit risk outcomes (Noriega et al., 2023). SVMs rank borrowers by finding the optimal hyperplane that separates creditworthy and risky categories (Ziemba et al., 2023). As an ensemble approach, random forests merge decision trees to capture nonlinear relationships, thereby improving robustness by going beyond traditional logistic regression in terms of predictive accuracy (Moscato et al., 2021).

However, challenges persist regarding transparency and regulatory compliance. Most ML algorithms operate as a so-called "black box" because credit decisions are difficult to explain to customers and regulators. This has inspired recent research into XAI techniques, bias-mitigation strategies, and fairness-aware algorithms to render credit scoring models as fair and accurate as possible, by following industry standards

(Kisten and Khosa, 2024). Fig. 3 shows the different factors impacting credit scoring, including conventional data sources and those used by non-banking entities to determine the creditworthiness of a borrower.

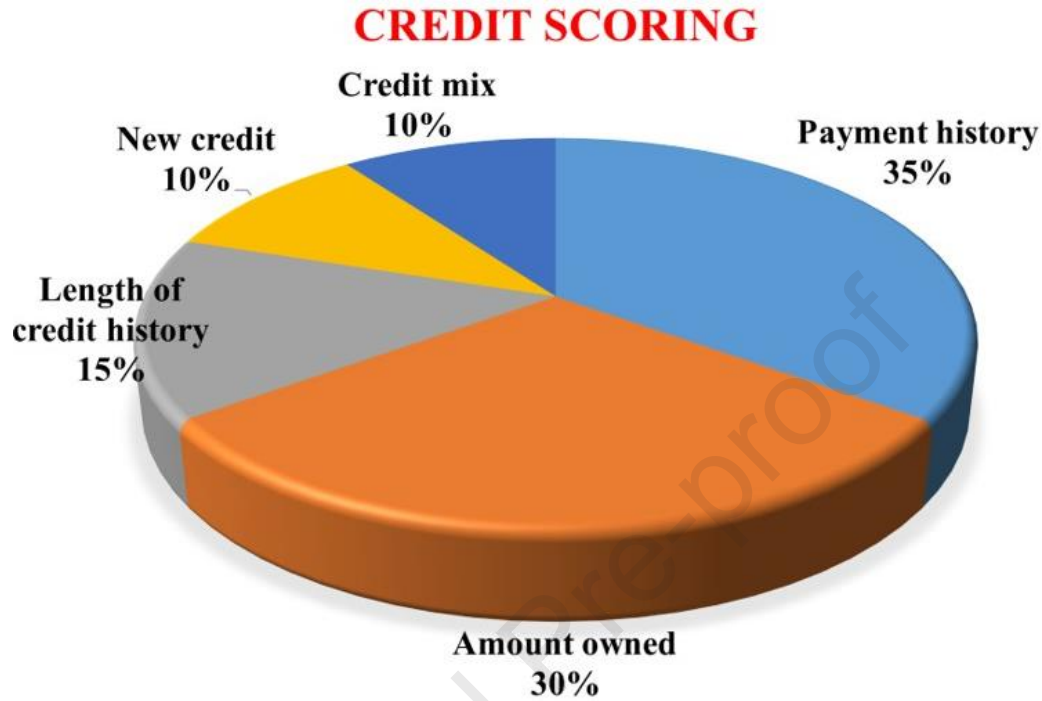


Fig. 3. Key components that contribute to credit scoring.

3. Quantum computing (QC): fundamentals

QC is a new and evolving paradigm that uses the laws of quantum mechanics to control phenomena at atomic and subatomic scales for information processing in a manner completely different from classical methods. Classical bits represent only distinct '0' or '1' states (Gill et al., 2022). By contrast, quantum bits or qubits can retain many states corresponding to superposition, allowing exploration of quantum states in parallel. This ability increases the opportunity for solving difficult problems in finance, such as portfolio fine-tuning and risk simulations. QC is particularly promising in finding solutions to complex, NP-hard problems, such as large-scale optimization and financial simulations, with improved precision and efficiency (Ajagekar et al., 2022). However, hardware constraints such as low qubit counts and high susceptibility to noise and operational errors have restricted the potential of current quantum systems, referred to as noisy intermediate-scale quantum (NISQ) devices (Bharti et al., 2022). Active research is being conducted on the development of error correction methods and the

construction of fault-tolerant architectures for quantum systems to outperform their classical counterparts in practical applications and achieve a quantum advantage.

3.1. Quantum mechanics: theoretical foundation

Quantum mechanics is a fundamental theory of physics that elucidates the behavior of matter and energy in atomic and subatomic dimensions and constitutes the foundation of quantum computation. Quantum mechanics is fundamentally probabilistic, whereas classical mechanics is based on deterministic principles. The power of quantum computation is underpinned by the main phenomena of superposition, entanglement, and interference, making it probabilistic in nature (Khan et al., 2024).

3.1.1 Superposition: In classical computing, a bit exists in either a ‘0’ or ‘1’ state. By contrast, a quantum bit (qubit) can exist simultaneously in a linear combination of both states (Khrennikov, 2021), expressed as $|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $|1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ in Dirac notation. The vector representation of qubits is $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ $\{\alpha, \beta \in \mathbb{C}\}$ where α and β are complex numbers, referred to as a probability amplitude. A single bit can represent two states, and n qubits can represent 2^n states simultaneously. Fig. 4 provides a basic representation of the qubits on a Bloch sphere (Filatov et al., 2024).

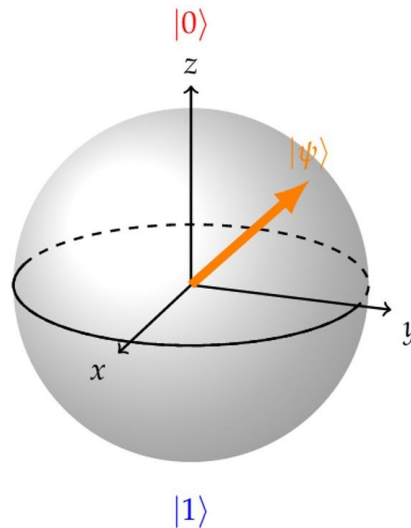


Fig. 4. Bloch sphere representation of a single-qubit quantum state. The vector demonstrates qubit superposition and phase in relation to the basis states $|0\rangle$ and $|1\rangle$ (Filatov et al., 2024).

Measurement of qubits in superposition collapses them into basic states with known probabilities. Unitary transformations enable constructive or destructive operations on qubits, thus enabling quantum parallelism following the principle of interference. Unitary transformations help preserve the norm of the vector and its inner product. The squared amplitude of the interference is used as the probability in QC. A unitary transformation can be expressed as $U U^\dagger = U^\dagger U = I$ where $U^\dagger$ is the Hermitian adjoint (conjugate transpose) of U . For a complex problem, the physical space required by classical computers grows exponentially with the amount of processing, in contrast to the linear growth experienced by quantum computers (S et al., 2022). This phenomenon is known as quantum parallelism, allowing quantum systems to explore many possibilities simultaneously, with exponentially increasing parallelism in computation. With this background, let us state that a classical bit is like a coin lying flat, with the heads or tails facing upwards. A qubit is similar to a coin spinning on its edges, thus simultaneously representing heads and tails until it is observed (Bansal and Kaur 2025). This ability to exhibit two possibilities simultaneously is known as quantum parallelism, the ability of quantum computers to evaluate many outcomes in parallel, thereby fast-tracking calculations.

3.1.2 Entanglement: The state of one qubit cannot be described independently of the others because of the nonclassical correlation between two or more qubits. Quantum cryptography and quantum teleportation depend on entanglement, enabling quantum gates to implement complex computations across multiple qubits simultaneously.

3.1.3 Interference: Interference is employed to increase the probability of the amplitudes of correct responses while suppressing those that are incorrect. This is a critical mechanism in algorithms such as Grover's search and quantum Fourier transform (Mundada et al., 2023).

3.2. From classical to quantum logic

Classical computation is predicated on irreversible operations that utilize logic gates, such as AND, OR, and NOT. For instance, an AND gate that generates an output of 0 does not indicate whether only one or both inputs are '0'. In contrast, quantum computation is inherently reversible because it is subject to unitary transformations governed by the Schrodinger equation (Herman, et al., 2023). This reversibility is

essential for preserving quantum coherence during computation. Nevertheless, measurements can induce irreversibility and collapse of the quantum state, thereby establishing a critical design constraint in the development of quantum algorithms.

3.3. Five stages of the quantum computing process

Quantum algorithms generally adhere to five conceptual states: (1) Qubits are utilized to convert a classical input into a quantum state; (2) Superposition is used to encode all conceivable states simultaneously; (3) The algorithm (oracle) is executed in parallel across all stages; (4) Quantum interference increases the probability of correct results, and (5) Qubits are measured to retrieve the classical output (Rietsche, et al., 2022). This structure is the foundation of quantum advantage, referring to the capacity to outperform classical systems in optimization, simulation, and computation tasks.

3.4. Quantum gate circuits

The quantum gates used to implement quantum computations are analogous to classical logic gates. This method employs unitary transformations to manipulate the qubit states. These entrances are classified into the X-gate (quantum NOT), Z-gate, and Hadamard (H-gate), which are single-qubit gates that generate superpositions. The controlled NOT (CNOT), Toffoli, and SWAP are multiqubit gates that are indispensable for the generation of entangled states and execution of logical operations across qubits (Mondal and Parhi, 2024). Quantum circuits are constructed by combining these gates and are employed to develop algorithms for real-world problems such as those in finance. Quantum gates, which are sequences of gates applied to qubits, are implemented in quantum circuits. These circuits are analogous to classical logic circuits but operate under quantum laws.

3.5. Quantum algorithms and their relevance to finance

Quantum algorithms offer substantial speedups over classical algorithms by utilizing the principles of superposition, entanglement, and interference.

- Grover's algorithm: Quadratic speedup is achieved in the search of unstructured databases. It has practical applications in fraud detection, credit assessment, and portfolio optimization.

- Shor's algorithm: Classical encryption systems are at risk because of their inability to efficiently factor large integers. The development of quantum-resistant cryptography has profound implications for financial cybersecurity (Rietsche, et al., 2022).
- Quantum phase estimation (QPE) and quantum Fourier transform (QFT): The core routines for simulating stochastic processes, QPE and QFT, are employed to evaluate the risks and pricing of derivatives (Sihare, 2025).
- Variational quantum algorithms (VQAs): Current NISQ-era devices are well suited for hybrid algorithms, including the quantum approximate optimization algorithm (QAOA) and variational quantum eigensolver. These are currently being investigated for credit risk assessment and option pricing selection (Wang et al., 2023).

3.6. Implementation challenges

Despite the potential of theoretical foundations, real-world QC encounters numerous obstacles (Chow, 2024):

- Hardware fragility: Qubits are exceedingly susceptible to noise and require exceptionally low temperatures to function.
- Scalability: Most real-world financial applications require extensive circuits and large qubit counts.
- Error corrections: Quantum error corrections are an ongoing development, and current systems are not fault tolerant.
- Algorithm maturity: Many quantum algorithms are still experimental and require additional development for implementation in financial environments.
- Cybersecurity risks: The potential exists for QC to render conventional encryption methods obsolete, thereby introducing new risks into digital finance systems.
- Ethical and regulatory factors: The disruptive nature of quantum advantages raises concerns in areas such as encryption, compliance, and transparency.

QC introduces a new paradigm that has the potential to substantially transform the financial sector by enabling high-speed computations, complex simulations, and secure communications. The integration of quantum hardware and hybrid quantum–classical strategies into financial systems is anticipated to revolutionize the way institutions model, forecast, and optimize operations as they continue to advance (Zhou,

2025). However, thoroughly leveraging these capabilities, makes consistent progress imperative in quantum algorithm development, hardware stability, and supportive regulatory frameworks. With these developments, quantum-enhanced finance is on the brink of becoming a fundamental component of the FinTech ecosystem of the future.

4. QML

4.1. *Integration of ML with QC*

QC and quantum information (QI) are gaining traction owing to their potential advantages. Recent advancements claim to have achieved quantum advantages in specific tasks. Quantum computers utilize sequences of quantum logic gates for processing (Padhi and Charrua-Santos, 2021). Qubits, which are the basic units of QC, can exist in superposition states. When qubits in superposition are measured, they collapse into basic states with known probabilities. Unitary transformation operations on qubits showcase quantum parallelism (Mahmud et al., 2020). This emerging technology can be integrated with ML, to improve the performance of ML algorithms and provide a method for developing a new field called QML, which merges the ML and QC concepts for enhanced solutions.

Using quantum computing in conjunction with conventional ML approaches, QML harnesses quantum phenomena, such as superposition and entanglement, allowing algorithms to perform vast data-processing tasks more efficiently than the classical ML, thereby opening up new possibilities for financial modeling and decision-making. ML typically entails teaching a computer to recognize patterns or make predictions from data. In classical ML, hardware problems prevent processing data points in large batches or in parallel. Conversely, quantum computers utilize quantum parallelism to encode and quantify multiple data points/features simultaneously, thus providing faster training and better accuracy, which is particularly advantageous for the complex datasets often considered in financial markets. Quantum parallelism, which allows countless scenarios to be evaluated simultaneously, provides an edge in problems such as fraud detection and market forecasting, where all possibilities need to be explored quickly. During data encoding, classical data are converted into quantum states using processes such as amplitude encoding. Quantum algorithms such as QSVM and quantum principal component analysis, adapt classical ML tasks by leveraging

quantum properties (Martín-Guerrero and Lamata, 2022). Hybrid models, which include hybrid quantum–classical models, divide workloads, whereby quantum systems handle tasks such as optimization or matrix computations and classical systems manage simpler processes (Fan and Han, 2022). Quantum-enhanced advanced techniques such as quantum clustering and quantum autoencoders exploit quantum mechanics to improve learning algorithms and computational speed (Wu et al., 2024). Using these techniques, QC and ML are integrated in QML.

4.2. QML: a new frontier

As a transformative frontier at the intersection of ML and QC, QML employs quantum phenomena, including superposition, entanglement, and quantum interference, to potentially surpass classical algorithms in terms of resource efficiency, accuracy, and speed, compared to classical ML, which is restricted by hardware limitations and algorithmic obstacles. Recent advancements in VQAs and NISQ devices have contributed to QML evolving from a theoretical curiosity to a pragmatic paradigm (Bharti et al., 2022). In high-dimensional data environments such as finance, drug discovery, and cybersecurity, this transition signals the beginning of a new era of computational innovation.

4.3. Quantum-enhanced ML models

The objective of quantum-enhanced ML models is to replicate or expand classical algorithms on quantum hardware. These models are not solely quantum renditions of the existing algorithms; they frequently incorporate entirely new computational strategies.

QSVMs: Quantum kernels are employed to embed data into high-dimensional Hilbert spaces, thereby enabling linear separation. These outperform classical SVMs in the detection of anomalies, classification of images, and identification of financial fraud. NISQ devices are scalable through hybrid quantum–classical frameworks and resilient to the noise in NISQ devices (Marengo and Santamato, 2025; Martín-Guerrero and Lamata, 2022; Zhou et al., 2017).

QNNs: Quantum operations are implemented to accelerate and simulate training on complicated manifolds. Quantum implicit representation networks and quantum convolutional neural networks have demonstrated enhanced performance in financial

forecasting and image super-resolution. Quadratic enhancements in decoding errors and up to 97% accuracy in credit scoring tasks have been demonstrated by QNNs (Martín-Guerrero and Lamata, 2022).

Quantum approximate optimization algorithm (QAOA): This is a hybrid variational approach for combinatorial optimization problems. Utilized in applications such as supply chain logistics, route planning, and portfolio optimization, it surpasses classical heuristics in terms of both the quality and efficiency of the solutions (Wang et al., 2023).

4.4. Tools and frameworks for QML

The advancement of robust libraries and tools has enabled QML to gain momentum. Classical ML operations that are seamlessly integrated with these platforms easily abstract quantum programming.

Table 1

Overview of leading tools and frameworks for quantum machine learning (QML).

Tools	Key features	Reference
Qiskit	International Business Machines Corporation's open-source quantum Software Development Kit.	Hevia et al. (2022)
PennyLane	Hybrid quantum-classical machine learning framework. integrated with PyTorch and TensorFlow.	Khan and Robles-Kelly (2020)
TensorFlow quantum	Enables quantum circuit simulation within TensorFlow; ideal for QNNs and QAOA.	Martin-Guerrero and Lamata (2022); Wang et al. (2023)

Note: QNN: quantum neural network; QAOA: quantum approximate optimization algorithm.

These tools facilitate the seamless integration of quantum algorithms into contemporary AI workflows and democratize access (Table 1).

4.5. QML algorithms vs classical ML counterparts

Compared to their classical counterparts, QML algorithms exhibit clear advantages in a variety of categories, including hardware dependency, accuracy, training efficiency, data-handling capacity, and computational speed. A comparison of the CML and QML approaches is presented in Table 2, with significant distinctions between them across important performance metrics (Khan and Robles-Kelly, 2020).

Table 2

Comparison of classical and quantum machine learning (ML) (Khan and Robles-Kelly, 2020).

Aspect	Classical ML	Quantum ML
Computation speed	Limited by classical CPU/GPU	Can offer exponential speedups in specific tasks
Data handling	High-dimensional data incur high costs	Efficient encoding and quantum parallelism
Model training	Gradient descent-based, sometimes slow	VQCs can converge faster
Accuracy in noisy data	Drops significantly	QNNs/QSVMs maintain accuracy in noise-heavy environments
Hardware dependency	Requires high-end GPUs	Runs on quantum processors (IBMQ, IonQ)

Note: VQC: variational quantum circuit; QNN: quantum neural network; QSVM: quantum support vector machine; IBMQ: IBM quantum; IonQ: ion quantum; CPU: central processing unit; GPU: graphics processing unit.

4.6. QML algorithms

Introducing QML algorithms without prior knowledge of QC can be difficult. These algorithms can be viewed as improved versions of classical ML models that use quantum features to handle data more effectively and reveal intricate relationships that traditional models may overlook, which is useful for financial experts. The following

subsections describe the main QML algorithms, along with a simplified interpretation of their financial applications. Combining QSVMs and evolutionary algorithms has enabled novel feature selection techniques that maximize classification accuracy while reducing the number of features. As evidenced by datasets on lung cancer, this method not only increases model efficiency but also identifies pertinent features that improve predictive performance (Arafa et al., 2023). QML algorithms combining quantum mechanical principles and ML techniques, have advantages over traditional approaches. Multiple states can be represented using QML algorithms, which allow parallel information processing. The correlation between quantum bits enabled by the entanglement feature facilitates learning by modeling intricate relationships more accurately (Roy et al., 2024). As QML can handle massive datasets more efficiently, it has the potential to completely transform the finance industry (Chavan and Mulay, 2024).

4.6.1. QSVM

The QSVM is a quantum-enhanced version of the well-known classical ML algorithm for classification and anomaly detection. By embedding the data into quantum space, where the patterns are easier to separate than in classical space, QSVM can classify complex, real-world financial data, such as fraud signals or credit risk, more accurately and faster than the classical SVM. The QSVM is a product of the intersection of the QC and ML domains. By leveraging quantum principles, QSVM improves the SVM performance. QSVM outperforms classical SVM in scenarios requiring high computational power, mainly in complex data environments. The quantum feature map in a QSVM plays a crucial role in the detection of complex patterns in high-dimensional data (Wang et al., 2024). Quantum feature maps allow the embedding of classical data into quantum states, thereby improving the exploration of complex relationships within the datasets. Using different quantum kernels, such as those derived from rotational gates, QSVM can achieve extraordinary classification performance (Abdulsalam and Ahmad, 2025). Quantum kernel techniques are used by quantum one-class SVMs to detect anomalies, combining randomized measurements and variables to improve performance. This approach allows the effective identification of outliers in large datasets, demonstrating the multi-nature aspect of quantum feature maps (Hidalgo, 2025). Tancara et al. (2024) introduced quantum kernels based on

ground-state manifolds and single-state tomography, demonstrating improved classification of dynamical singularities in multi-qubit systems.

The QSVM models are resilient to noise and maintain performance, even in near-term quantum devices (Suzuki et al., 2024). Lee and Park (2024) evaluated different Pauli configurations (ZZ, ZY, and a custom configuration) and found that these configurations significantly affect execution accuracy in binary classification tasks. QSVMs utilize quantum parallelism, allowing them to process multiple data points simultaneously, which can significantly reduce computation time compared to classical SVMs (Ramesh et al., 2024). Combining QSVMs and evolutionary algorithms has enabled novel feature selection techniques that maximize classification accuracy while reducing the number of features utilized in combination.

4.6.2. Quantum encoders

Quantum encoders play a critical role in quantum error correction and are essential tools for reliable QI transmission and processing. Quantum encoders accomplish this by mapping logical qubits to physical qubits to improve noise and decoherence resistance. Current developments in quantum encoders focus on fault tolerance and efficiency issues, which are crucial to the success of quantum computers in practice (Hayashi and Wang, 2022). Quantum low-density parity-check codes are designed to minimize error propagation using multi-qubit gates. These encoders utilize shared entanglement and group qubits in contiguous blocks, allowing transversal implementations that enhance fault tolerance. The primary goal of these encoders is to reduce error propagation that can occur during quantum computations (Bausch et al., 2024). Efficient recursive encoders represent a major advancement in quality risk management code construction, focusing on the optimization of resource use with minimal impact on the performance level. The encoders are resource efficient as they achieve a circuit depth of $O(\log n)$ (Borrego-Carazo et al., 2020). The quantum circuit autoencoder compresses the information inside quantum circuits for anomaly detection and noise mitigation. It employs a variational quantum algorithm to achieve effective compression and recovery fidelity (Wu et al., 2024). Quantum encoders may be promising for improving quantum computation; however, achieving universal applicability across a variety of quantum systems and maintaining performance under varying conditions, still pose a challenge.

4.6.3. QNN

As a quantum generalization of classical neural networks, QNNs harness qubit-based quantum computation to represent intricate data structures. Financial professionals regard QNNs as powerful models capable of uncovering subtle patterns in noisy market data, thereby capitalizing on them for credit default and stock price forecasting. QNNs represent the intersection of QC and ANNs, leveraging quantum mechanics to enhance computational capabilities (Jia et al., 2019). These networks, which can be structured similarly to classical neural networks, utilize quantum bits (qubits) and quantum operations, potentially offering superior performance in various tasks. QNNs can be used to represent complex data structures. As evidenced by their improved performance on datasets such as the Modified National Institute of Standards and Technology, QNNs may perform better than conventional networks in applications such as picture categorization (Wang et al., 2022). QCNNs demonstrate a remarkable reduction in convergence time, achieving results in just 50 iterations, compared with over 200 iterations for their classical counterparts (Gong et al., 2023). QNNs provide a quadratic improvement in the decoding errors for QI, thereby enhancing the reliability of processing under noise (Nguyen et al., 2020). Quantum implicit representation networks are specialized in representing high-frequency signal components. They perform better than traditional NN in image super-resolution and generation tasks (Ma et al., 2025), whereas QNNs have demonstrated superior accuracy in tasks such as credit default prediction, achieving a prediction accuracy of 97.63% compared with 91.20% for classical networks (Sengar et al., 2023). In time-series forecasting, QNNs can effectively model signals with high noise levels, outperforming classical models such as bidirectional long short-term memory (BiLSTM) which requires more parameters. QNNs can be trained significantly faster than classical ML models, which is crucial in the fast-paced financial environment, where timely decisions are essential (Zhou et al., 2025).

4.6.4. Quantum clustering

Quantum clustering operates based on quantum properties and is more efficient than classical clustering algorithms in organizing financial data. Portfolio diversification and risk management rely on identifying clusters of assets and the dynamic correlations among them. As an improvement on conventional clustering

techniques, quantum clustering is a novel data analysis method based on the concept of QC. This method is particularly useful for high-dimensional datasets, where traditional algorithms frequently struggle with accuracy and efficiency. To overcome the limitations of traditional clustering techniques, quantum clustering algorithms such as Quantum Local Unitary Embedding (qLUE) use quantum mechanics to achieve linear scaling in both the number of data points and their density, thereby addressing the challenges faced by classical clustering methods (Qi et al., 2024). Quantum clustering uses density-based methods to find cluster centers through potential functions, thereby improving the identification of data classes without any prior labels (Bhattacharjee and Mitra, 2020). QC is based on the Schrödinger equation, and data clustering is achieved by identifying the extremes of a potential function (Nabatian et al., 2023). QA improves clustering by formulating the problem in a quadratic binary optimization framework, thereby enabling the discovery of global minima more effectively than classical methods (Jiang and Chu, 2023).

Quantum clustering aids in dimensionality reduction, allowing smooth visualization and understanding of complex data (Thompson and Siopsis, 2023). Dynamic quantum clustering, which differs significantly from traditional methods such as density-based spatial clustering of applications with noise and k-means, has shown excellent performance in the stock market. These techniques adapt better and provide better results in dynamic stock market environments. An improved analysis helps investors make better decisions, potentially increasing the accuracy of their investment strategies (Das et al., 2024a, 2024b). Eigen portfolios are used to analyze the correlation matrix of the financial assets. This technique helps identify clusters of assets, which is crucial for effective portfolio diversification and risk management. The better understanding provided by quantum clustering also addresses the Epps effect which indicates that the correlation between assets changes over time (Narwane et al., 2025). Despite the potential of quantum clustering, it is essential to consider the limitations of current quantum technologies, in that, it may not yet provide significant speed-ups in all applications, particularly in real-world scenarios such as optical-fiber communication (Bhaskaran and Prasanna, 2025).

5. QML in financial applications

5.1. Introduction to QML in finance

QML is applied in finance across multiple domains, including risk assessment, credit scoring, market simulation, high-frequency trading, fraud detection and anti-money laundering, portfolio optimization, and option pricing, forming an integrated framework for advanced financial decision-making. In the financial sector, QML offers a paradigm shift by leveraging the ability of QC to handle high-dimensional data spaces, complex optimization problems, and stochastic simulations. In contrast to CML, QML algorithms have the ability to employ quantum parallelism to accelerate a variety of activities, such as credit scoring, fraud detection, risk modeling, and portfolio optimization by tenfold (Beer et al., 2020; Marengo and Santamato, 2025; Zhou et al., 2017). Although the use of quantum algorithms in financial analytics is relatively new, investigations based on simulations and experiments have yielded encouraging results. This section examines the fields in which QML has advanced rapidly, as evidenced by cutting-edge quantum algorithms and new experimental findings.

5.2. QML framework for financial applications

Outlining the typical QML workflow in finance is crucial prior to exploring particular areas. The relationship between data pretreatment and model performance assessment can be clarified by this organized approach (Lamichhane and Rawat, 2025). Using amplitude encoding, angle encoding, or a combination of the three, traditional financial data are first turned into quantum states in a QML workflow. Then, based on the type of problem and the features of the data, the right QML algorithms are chosen. These could be QSVMs, QNNs, or the QAOA. During the training process, hybrid quantum-classical loops are often used to optimize quantum circuits with variable parameters so that the chosen loss function is as small as possible. Finally, the quantum states are measured, classical information is extracted through post-processing, and the model's success is judged by things like F1-score, accuracy, and root mean square error (RMSE).

5.3. Application domains

5.3.1. QML for fraud detection and AML

A frequent application of QML is fraud detection. QML significantly facilitates the detection of irregularities and fraudulent patterns in large unbalanced datasets. For instance, quantum variational classifiers and QNNs exhibit significantly better performance in supervised learning tasks (Beer et al., 2020; Marengo and Santamato, 2025; Martín-Guerrero and Lamata, 2022; Zhou et al., 2017). Such classifiers are used to detect fraudsters through the entanglement and superposition present in quantum features. These models can improve classification accuracy through quantum-enhanced pattern recognition. Because QSVM utilizes quantum kernels to map classical data into very high-dimensional quantum states, it registers a greater precision and F1-score than classical techniques (Qi et al., 2024; Abdulsalam and Ahmad, 2025; Lee and Park, 2024). In addition, QGNN provides a more instrumental framework for detecting complex patterns of fraud in network financial data. According to the results, QGNNs have a fraud detection accuracy of 94.5%, which is higher than the 92.3% of traditional GNNs (Innan et al., 2024). The efficiency and accuracy of QGNN can be further increased by combining it with variational quantum circuits (VQCs).

Several empirical studies have supported these findings. Innan et al. (2024) studied quantum support vector classifiers (QSVC), variational quantum classifiers, estimator QNN (EQNN), and sampler QNN (SQNN) models for financial fraud detection using the BankSim dataset. Among the various feature maps, QSVC with ZFeatureMap had the highest performance with an F1-score of 0.98, whereas EQNN and SQNN performed poorly because of their simpler circuit structure and related issues with high-dimensional data. Kyriienko and Magnusson (2022) designed a quantum anomaly detection scheme for CCF claiming quantum kernels with data-driven rotations, variational circuits, and Hamiltonian evolutions. This scheme performed better than classical kernels using radial basis function, with gains of over 10% in average precision in a supervised task and over 15% in an unsupervised task, demonstrating the advantage that faster quantum hardware can provide to fraud detection efficacy.

The quantum-enhanced synthetic minority oversampling technique (SMOTE) has also been used by QML to treat the class imbalances in quantum Boltzmann

machines (QBM)s developed for fraud detection. QBMs are theoretically capable of identifying abnormal features that may indicate illegal activity. This capacity is particularly relevant to AML, in detecting rare behavioral transaction patterns (Oztas et al., 2024). Mancilla and Pere (2022) also analyzed the role of linear discriminant analysis (LDA) and found it to be crucial for improving the results of VQAs and QSVCs in CCF detection. LDA seeks to achieve maximum class separability and hence provides better results than principal component analysis (PCA), singular value decomposition (SVD), and supervised kurtosis projection pursuit in terms of balanced accuracy, recall, and F1-scores; however, the precision is marginally lower. This highlights the importance of efficient preprocessing for maximizing the potential of quantum algorithms in supervised fraud detection. Hybrid quantum–classical models provide better feature selection, reduce feature dimensionality, and improve fraud detection accuracy.

Grossi et al. (2022) noted that a hybrid QSVM implemented using IBM Qiskit performed better than classical methods in terms of accuracy and false-positive rates in financial fraud detection. With the integration of the quantum feature importance selection algorithm, feature selection and dimensionality reduction perform better than traditional methods such as factorial analysis of mixed data, making quantum fraud detection more useful. However, issues, such as model scalability, algorithm complexity, and the limits of quantum hardware hinder their technological potential. Nevertheless, QML is anticipated to become a critical component of real-time fraud detection and crime prevention strategies in finance as QC technology advances. Although fraud detection and AML are currently the most established QML use cases in finance, this technology has much broader applications.

QML in insurance actuarial sciences: QML models can speed up computationally complex actuarial calculations through quantum parallelism in stochastic simulations of life tables, frequency-severity models, and risk capital estimation. For example, quantum amplitude estimation can reduce the computational burden of Monte Carlo simulations used for solvency calculations, enabling insurers to process more granular demographics and claims data in near-real time (Amjad, 2025). VQAs can also optimize premium pricing structures by analyzing high-dimensional policyholder data while capturing nonlinear dependencies that classical models may miss.

QML for derivative pricing and risk management: Quantum algorithms such as quantum amplitude estimation (QAE) and quantum Monte Carlo have demonstrated the ability to price complex path-dependent derivatives (e.g., Asian, barrier, and basket options) with a quadratic speedup over classical Monte Carlo methods. QML can also enhance portfolio-level risk simulations by modeling tail dependencies in correlated assets more accurately, thereby better supporting value-at-risk and conditional value-at-risk estimation. These capabilities are particularly relevant in volatile market environments, where classical simulation runtimes can become prohibitive (Zhou, 2025).

By incorporating these broader financial contexts—insurance actuarial modeling, derivative pricing, and portfolio risk assessment—the scope of QML in finance becomes clearer. Although real-world implementations remain constrained by NISQ-era hardware limitations, hybrid approaches and ongoing hardware advances are steadily moving from academic proofs of concept to production-ready solutions.

5.3.2. *QML for portfolio optimization and algorithmic trading*

Portfolio optimization involves selecting the optimal mix of assets to maximize returns while minimizing risk, which is typically modeled as a quadratic optimization problem (Zhou, 2025). Although CML-based methods such as gradient descent and evolutionary algorithms offer satisfactory performance, they can become computationally prohibitive when scaling to large asset universes with complex constraints. QML-based approaches such as QAOA can process vast solution spaces more efficiently. These methods exploit quantum tunneling to escape the local minima and offer potential advantages in global optimization. However, real-world deployment requires handling NISQ devices and integrating quantum outputs with existing financial systems. As portfolio optimization for strategic considerations is inherently time insensitive, this process is an early candidate for taking advantage of the QML approach; however, robust error mitigation strategies are required to be truly useful. Rosenberg et al. (2015) examined binary, unary, sequential, and partitioning encoding schemes under D-Wave's quantum annealer for multiperiod portfolio optimization problems formulated as quadratic unconstrained binary optimization (QUBO). They observed that unary encoding is more robust for large integers, and binary encoding is the fastest without noise tuning. Compared with classical solvers, embedding

improvements and higher success rates on noise-tuned devices were observed, highlighting the promise of QA along with the need for further hardware enhancement.

To solve the Markowitz problem of portfolio optimization on the Abu Dhabi Securities Exchange, Elsokkary et al. (2017) adopted an adiabatic quantum optimization approach on a D-Wave processor and efficiently used QUBO-Ising mapping to tradeoff risks and rewards. Using reverse annealing on a D-Wave 2000Q processor, Venturelli and Kondratyev (2019) offered a hybrid approach for portfolio optimization that starts with classical solutions and uses pauses to further investigate local minima. The described algorithms are better than forward annealing and genetic algorithms, with speedups ranging from 10 to 1,000 times for intermediate problems involving 24 to 60 assets. Hybrid methods use classical solutions to improve the efficiency of the quantum search in large-scale optimization problems. QA attempts to use quantum fluctuations to find low-energy states and has difficulty in finding widely separated minima, whereas QAOA tries to constructively use quantum interference to deterministically solve problem classes that have otherwise exponentially small success probabilities. Algorithms such as Harrow–Hassidim–Lloyd (HHL) and second order cone programs (SOCP) provide notable runtime improvements in scalable portfolio management and risk-return analysis. Rebentrost and Lloyd (2024) proposed a quantum portfolio optimization algorithm that exploits HHL for a fast risk-return trade-off analysis, with exponential or polynomial speedups in matrix inversion. Kerenidis et al. (2019) developed a quantum algorithm for constrained portfolio optimization that achieved an $O(n)$ speedup by reducing the problem to SOCP and applying the quantized interior-point method, which resulted in both efficiency and runtime gains in complex financial optimization tasks.

Algorithmic trading demands extremely low-latency decision-making, in which even microsecond delays can translate into significant financial losses (Maronese et al., 2024). Although QML offers theoretical speedups in pattern recognition and predictive modeling, the current hardware limitations and qubit decoherence times present challenges for meeting the sub-millisecond execution requirements. Hybrid quantum–classical architectures have been explored for pre-training models using quantum-enhanced data processing before deployment in a high-frequency trading (HFT) environment. The unique constraint is that the models must operate in real time without relying on quantum computations during the execution phase, making offline quantum

training with classical deployment a more feasible strategy. This highlights the importance of scenario-specific QML adaptation, whereby latency-critical operations remain classical, and quantum resources are used for offline learning and strategy optimization.

5.3.3. Risk assessment and credit scoring

The QNNs and QSVMs have proven to be highly effective QML models for evaluating various financial risks including operational, credit, and market risks. These quantum models can be used to analyze high-dimensional data and capture the complex interrelationships among the risk factors more efficiently than traditional methods. QML models often demonstrate superior generalization capabilities, particularly in scenarios with limited training data. For example, simulated risk analyses have shown that quantum-enhanced models forecast value-at-risk metrics with greater precision than classical DL networks (Zhu et al., 2023). QML techniques are becoming increasingly preferred in credit risk modeling because of their high-dimensional processing capacity and the volume of nonlinear data. Some implementations based on quantum k-nearest neighbors and QSVM have enhanced classification accuracy and reduced false positive rates (Zegundry et al., 2024). Experiments on synthetic and real-world datasets with the suggested method led to improvements in F1-scores between 5% and 7% when compared to conventional methods.

In supporting feature selection, QA becomes a better tool in credit scoring through redundancy elimination while retaining accuracy. Milne et al. (2017) used QUBO in conjunction with quantum annealers to solve the feature selection problem for credit scoring, producing optimal subsets of 24–35 features from German Credit Data with 0.76 ± 0.05 accuracy. By equalizing feature influence and independence, this method minimizes redundancy and complexity, thereby preserving accuracy and improving scalability compared to the classical recursive feature elimination method. For credit risk analysis, Rath and Date (2025) proposed a hybrid quantum classical deep neural network for row-type dependent predictive analysis (HyQuC-DeepNN RTDPA) along with a hybrid quantum–classical DL framework with CNN layers for quantum feature encoding. Using SMOTE and PCA, the model attained 83.47% accuracy for personal loans and 81.12% for agricultural loans, indicating strong generalization and an ability to detect standard and doubtful classes.

Taking advantage of these quantum layers under hardware and imbalance constraints, this approach can be used for fraud detection and portfolio risk forecasting, which greatly enhances the recognition of complex fraud cases. Thakkar et al. (2024) further developed random forests for churn prediction using deterministic determinantal point processes, showing a 6% precision improvement. QNN architectures with orthogonal and compound layers can achieve credit risk assessment performances comparable to their classical counterparts, with far fewer parameters. This approach improves sampling and training efficiency while addressing the missing data through advanced imputation and preserves accuracy with reduced computational resources. Leclerc et al. (2023) applied a neutral-atom quantum processor for fallen-angel forecasting, employing the quantile boost algorithm with QUBO and random graph sampling to optimize heterogeneous learners. The quantum classifier achieved a precision of 27.9% and recall of 83%, comparable to random forest benchmarks, but with fewer learners, shorter runtimes, and improved scalability via tensor network optimization. This enhanced performance is particularly beneficial for default prediction and subprime credit evaluation, areas where classical models often underperform.

5.3.4. QML for option pricing and market simulations

Computational efficiency is important in option pricing, especially in complex or exotic pricing. QAE, also called the fundamental quantum algorithm, provides a quadratic speedup with respect to classical Monte Carlo estimation (Maronese et al., 2024). In simulation models, the Black-Scholes and Heston models benefit vastly from QAE reduction in the number of simulations required for precise pricing. Ramos-Calderer et al. (2021) proposed a quantum algorithm for European option pricing based on the Black-Scholes model, using unary encoding to reduce circuit complexity and increase noise resilience in NISQ devices through amplitude distribution, payoff calculation, and amplitude estimation. Although better in error mitigation strategies, challenges such as qubit limitations and gate errors remain. Stamatopoulos et al. (2020) developed a gate-based quantum algorithm using amplitude estimation to obtain a quadratic speedup over classical Monte Carlo methods, successfully pricing barrier and multi-asset options by implementing error mitigation on IBM's Q Tokyo device. Sakuma (2020) proposed deep QNNs (DQNNs) for learning implied volatilities and option prices, accurately predicting call prices and Greeks using a rather simple

quantum architecture, thereby demonstrating sound risk management under the Black-Scholes framework. Bank dynamics which are faster, necessitate faster reaction times and therefore, more flexible and dynamic pricing mechanisms.

Recent simulations in Qiskit Finance demonstrate that QML-based pricing requires significantly less processing power than traditional approaches. To efficiently encode classical data distributions into quantum states, Zoufal et al. (2019) developed a framework for quantum generative adversarial networks (QGANs), which combines a quantum generator with a classical discriminator. When applied to the pricing of financial derivatives, this gives rise to quantum amplitude estimation, which is almost twice as fast and accurate as conventional Monte Carlo methods and thus opens up numerous opportunities for complex quantum finance algorithms. For the purpose of modeling financial data, Ganguly (2023) investigated QGAN and the quantum circuit born machine (QCBM) models. QCBM estimates distributions using optimizers such as *constrained optimization by linear approximation* and simultaneous perturbation stochastic approximation, whereas QGAN learns bitcoin pricing data with high fidelity. Both models demonstrated an ability to improve forecast accuracy and lower uncertainty in financial modeling activities. To advance quantum ML, Dutta et al. (2024) created a quantum attention deep Q-network (QADQN), which supports financial market prediction decision-making by employing quantum multihead self-attention mechanisms and VQCs. When QADQN was evaluated on S&P 500 data, it produced a variety of positive returns in both overlapping (75.37%) and non-overlapping (66.82%) periods. The Sharpe and Sortino ratios, which measure risk-adjusted superior performance, demonstrated QADQN's ability to mitigate downside risk.

Quantum-enhanced RL in the guise of VQC within QADQN provides a more adaptive and efficient mode of decision-making in financial markets. QNNs and generative models have been exploited to obtain market predictions and financial forecasting with increased accuracy, noise tolerance, and computational efficiency. However, challenges include quantum circuit noise, assertive scalability, and training hurdles. Future research should focus on making models more robust and scaling them up for wider real-world financial market applications. In addition to pricing, quantum algorithms such as VQAs and quantum walk are used to imitate, with a reasonably high

degree of fidelity, complex market dynamics, and behaviors that defy classical simulations. These quantum models capture investor behavior, market microstructure, and stochasticity at the granular level. Specifically, quantum walks have proven effective in modeling tail risk events, enabling speculators and risk managers to predict rare but significant market occurrences more accurately (Chang et al., 2024b). Additionally, quantum-based market simulations support the development of more resilient stress-testing scenarios, thereby improving regulatory compliance and risk-forecasting capabilities.

5.3.5. QML for high-frequency trading and strategy optimization

Quantum reinforcement learning (QRL) is a powerful and innovative approach that has been developed specifically for HFT. Quantitative trading is a fringy area in finance concerned with algorithmic trading, which analyzes historical market data and is amenable to complex trading algorithms. Statistical arbitrage is a common quantitative approach used by most hedge funds (Arthur et al., 2025; Narayan, 2019). The core concept of arbitrage is to exploit price discrepancies arising from the co-movement of securities and portfolios, to profit from other traders' pricing errors. Although numerous classical algorithms for quantitative trading have been proposed (Ezugwu et al., 2022; Jensen and Iosifidis, 2023; Yang et al., 2023; Zhang et al., 2019), and traditional hardware solutions such as infrared communication and field-programmable gate arrays have been employed in recent years (Cohen, 2022; Khan et al., 2025), these methods still struggle to meet the extreme speed requirements of HFT. In particular, HFT demands ultra-low latency and rapid computation to capitalize on fleeting opportunities (Mosavi et al., 2020).

In statistical arbitrage, identifying potential co-integrated pairs requires numerous linear regressions and co-integration tests on massive historical datasets. For instance, in US stock markets, the problem size can exceed $N = 10^7$ with a computational complexity of approximately 10^{15} , which makes it challenging for classical computers to handle efficiently. QC is a promising approach for tackling large-scale problems. In addition to processing transaction-level data with high granularity, QRL agents can analyze individual transactions, market updates, and limit orders. This enables them to execute optimal trading strategies with minimal latency and adapt swiftly to market changes. Research has demonstrated that in time-sensitive financial environments,

QRL agents outperform classical RL agents in terms of accuracy and flexibility (Butt et al., 2025). Enabling real-time modeling of the market microstructure, QRL is a valuable tool for institutional traders.

5.4. Case studies and experimental results

Recently, experimental results have illustrated the potential and viability of QML in real-world financial applications (Schetakis et al., 2022).

Recent breakthroughs underscore the increasing implementation of QML in financial applications by prominent technological companies and financial organizations. IBM Quantum, via its Qiskit Finance module, has deployed quantum techniques including the QAOA and QAE to enhance investment portfolio optimization and the pricing of intricate derivatives. JPMorgan Chase has explored the application of QML for fraud detection and AML solutions, whereas D-Wave Systems has created hybrid quantum-classical solvers for limited portfolio optimization. SandboxAQ has utilized QML methodologies to improve credit evaluation and risk analytics, while Google's TensorFlow Quantum framework has investigated QNNs for financial sentiment analysis and credit default forecasting. These case studies collectively illustrate that QML is swiftly evolving from theoretical inquiry to practical application, with many institutions acknowledging its capacity to revolutionize the financial sector.

5.5. Comparative analysis of recent QML research in finance

Recently, the financial industry has paid close attention to QML because of its ability to solve complex computational problems more efficiently than conventional techniques. The incorporation of quantum algorithms into financial applications, including time-series forecasting, derivative pricing, fraud detection, and credit risk assessment, has been the subject of numerous studies. However, this field is still in its infancy with significant gaps in hardware limitations, algorithmic optimization, and scalability. Table 3 presents a comprehensive comparative analysis of significant QML studies, emphasizing the research gaps, advantages, and algorithms employed while also identifying the challenges. This analysis emphasizes the practical constraints and potential of QML in real-world financial contexts.

Table 3

Literature review matrix for quantum machine learning (QML) applications with identified gaps and challenges.

Reference	Research gap	Advantage	Algorithm used	Challenges
Mironowicz et al. (2024)	Deeper exploration of QML required in quantum finance.	Provides a comprehensive overview of QML applications in quantitative finance.	- QA; - Variational quantum eigensolver; - QENN; - Quantum least squares Monte Carlo.	Lack of quantum error correction, noise, and scaling limitations in quantum computers.
Emmanoulopoulos and Dimoska (2022)	Need for quantum optimization in QNN training on large datasets.	Demonstrates effective time-series forecasting using parameterized Q circuits.	- Parameterized quantum circuits as QNNs; - Bidirectional long short-term memory neural networks.	Limited data volume for QNN training impacts model accuracy.
Alcazar et al. (2022)	Innovation is required in quantum circuit design beyond Monte Carlo methods.	QML significantly improves credit valuation adjustment calculations.	- Credit valuation adjustments; - Bayesian variant of quantum amplitude estimation; - K-qubit unitary operator.	Requires efficient circuit design due to NISQ-era limitations.
Tang et al. (2022)	Need for custom quantum algorithms tailored for financial use.	Highlights QML's potential to boost financial	- QAOA; - Quantum adiabatic computation;	Quantum device scale restricted; Lack of generalized quantum

		computation efficiency.	- Quantum Monte Carlo Method; - QNNs.	algorithm frameworks.
Dutta et al. (2024)	Inadequate testing of QADON and quantum circuit challenges in RL.	QADQN surpasses traditional trading strategies and buy-and-hold approach.	- Quantum deep; - Q-network;	Quantum decoherence, circuit scalability, and implementation hurdles in RL frameworks.
Xu et al. (2024)	Exploration of hybrid methods in time-series prediction limited.	QGAF method enhances accuracy in financial time-series forecasting.	Quantum genetic algorithm	Overfitting in standalone ML models; Requires better sample handling.
Grossi et al. (2022)	Techniques for building quantum kernels limited in real-world classification.	Quantum and classical methods combined for improved classification.	- QSVM; - XGBoost; - Random forest.	Q-hardware constraints and challenges in handling large datasets.
Rebentrost et al. (2018)	Integration of QC into financial modeling requires further exploration.	QA achieves quadratic speedup in the Monte Carlo pricing of financial derivatives.	Quantum Monte Carlo and amplitude estimation algorithm	Maintaining high confidence in price estimates and leveraging speedup on real hardware.
Doosti et al. (2024)	Additional real hardware experimentation required for financial QML.	Highlights the potential of QML in credit scoring, fraud detection, and risk management.	Quantum-enhanced classical models (generalized).	Data loading inefficiencies and challenges in training QML models on quantum hardware.

Note: QA: quantum annealing; QENN: quantum Elman neural network; QNN: quantum neural network; NISQ: noisy intermediate-scale quantum; QAOA: quantum approximate optimization algorithm; QADQN: quantum attention deep Q-network; RL: reinforcement learning; ML: machine learning; QSVM: quantum support vector machine; QC: quantum computing; QGAF: Quantum gramian angular field.

6. Challenges in classical and quantum ML for financial applications

6.1. Challenges in CML

Despite its transformative potential, CML in finance continues to encounter numerous constraints that inhibit its extensive implementation in mission-critical financial systems. These obstacles include cybersecurity vulnerabilities, algorithmic complexity, regulatory compliance, and data quality.

Data quality and availability: The availability of clear, structured, real-time data is a critical component of financial ML models. Noise, missing values, and latency are frequently encountered in market data and can result in model failure which distorts predictions (Vancsura et al., 2025).

Non-stationarity and market volatility: Financial time series are inherently nonstationary and exhibit heteroscedasticity, autocorrelation, and structural discontinuities, which present challenges to the generalization of ML models.

Overfitting and interpretability: Although DL models are highly effective, they are frequently susceptible to the overfitting of historical data and lack transparency. In environments with high consequences such as credit scoring and risk management, explainability is crucial for stakeholder trust and conformance.

Cybersecurity risks: Integrating ML into financial systems creates new attack surfaces. Data vulnerabilities, adversarial attacks on models, and model manipulation are urgent concerns.

Regulatory compliance: Interpretability, auditability, and impartiality of ML models require stringent regulations such as the EU general data protection regulation,

Basel III, and Securities and Exchange Commission guidelines. These features are difficult to achieve using traditional black box models (Nastoska et al., 2025).

Efforts to address these challenges involve the development of hybrid models that integrate statistical and ML approaches, utilization of XAI frameworks, and implementation of robust validation techniques (Machlev et al., 2022; Minh et al., 2022). Nevertheless, as financial systems become increasingly intricate, the scalability and optimization constraints of classical ML are being addressed through novel methods that include quantum computation.

6.2. Challenges in QML

Although QML provides theoretical advantages over classical models, particularly in high-dimensional data processing and optimization, its practical application in the financial sector has been hindered by numerous obstacles (Ranga et al., 2024):

Hardware limitations: Decoherence, gate errors, and qubit connectivity issues are prevalent in current quantum hardware (NISQ devices). These constraints limit the depth and run time of the algorithms, rendering them impractical for performing lengthy and intricate quantum computations.

Data encoding bottlenecks: The process of efficiently importing classical financial data into quantum systems is complex, with amplitude encoding and quantum feature maps necessitating a compromise between feasibility and accuracy.

Algorithmic maturity: The development of promising quantum algorithms such as QSVM, QAVA, and QNN is still in its early phases, and widespread benchmarking against classical models on financial datasets is lacking (Ullah and Garcia-Zapirain, 2024).

Scalability issues: The exponential cost of simulation on classical hardware and the quantum resource limitations of numerous QML algorithms render them incapable of scaling efficiently.

Security and ethics: In the future, quantum algorithms may disrupt existing cryptographic systems, thereby posing substantial ethical and legal concerns. Financial systems require quantum-safe protocols.

Regulatory readiness: Financial regulations have not yet been developed to accommodate the stochastic behavior and black-box nature of quantum models. QML adoption may have been impeded by the absence of a governance framework.

Overcoming these obstacles requires implementing regulatory innovations that correspond to the growth rate of quantum technology, quantum error correction methods, and robust hybrid architectures. QML is steadily gaining traction in the financial industry because of its ability to handle complex, high-dimensional data and complete optimization tasks more quickly than CML. Table 4 compares CML and QML in financial applications, focusing on the advantages and disadvantages of the technology (Devadas and T., 2025; Innan et al., 2024; Khan and Robles-Kelly, 2020).

Table 4

Comparative analysis: classical machine learning vs. quantum machine learning (QML) in finance (Devadas and T., 2025; Innan et al., 2024; Khan and Robles-Kelly, 2020).

Aspect	Classical ML	Quantum ML
Performance benchmarks	Demonstrates efficacy in real-world applications, including trading, credit assessment, and fraud detection.	Limited benchmarks: although small-scale evaluations indicate potential, consistent superiority has not been demonstrated.
Resource and computational complexity	Requires substantial classical computing resources, such as GPU/TPU. High-dimensional optimization can become unfeasible.	Quantum algorithms offer exponential speedup for specific tasks (e.g., QAOA for portfolio optimization); however, hardware constraints prevent this performance in practice.
Interpretability and	Tools for model explication that are mature (e.g., SHAP,	QML is primarily a black box. Quantum models are still in the

explainability	LIME). Crucial for organizational compliance.	early stages of developing interpretability techniques.
Scalability and infrastructure	Utilizes distributed computing to effectively handle large amounts of data.	Quantum error rates and quantity of stable qubits are limiting factors. Scalability is still a significant challenge.
Maturity and real-world readiness	Cloud providers and open-source tools provide robust support for the widespread adoption of FinTech applications.	Remains primarily experimental. Continuous research is being conducted to apply theoretical concepts to practical applications.

Note: ML: machine learning; QAOA: quantum approximate optimization algorithm; SHAP: Shapley additive exPlanation; LIME: local interpretable model-agnostic explanation; GPU: graphics processing unit; TPU: tensor processing unit.

6.3. Challenges in the real-world implementation of CML and QML in finance

The implementation of AI-driven models in live financial environments necessitates surmounting significant real-world obstacles in both classical and quantum paradigms (Majid et al., 2025):

Data availability and quality: High-quality, labeled, and timely data are indispensable; however, they are not consistently accessible, particularly that quantum applications require low-noise datasets.

Quantum hardware limitations: The fragility of quantum devices is a constraint on QML. The hardware currently used is either insufficient or intended for large-scale production.

Algorithm scalability: The increasing complexity of financial problems make it imperative to evaluate the scalability and robustness of classical and quantum algorithms.

Regulatory and ethical issues: Black box models are a source of concern for regulated industries. Financial AI governance continues to be fundamentally concerned with algorithmic bias, model accountability, and auditability.

Integration with existing systems: Significant infrastructure investment is required to integrate legacy systems into institutions and trading platforms as they are frequently incompatible with contemporary CML/QML frameworks.

7. Future scope

7.1. Future scope in ML

Algorithmic trading, risk modeling, fraud detection, and personalized customer service are all being transformed by ML, which continues to revolutionize the financial sector. Advancements in multimodal ML systems, RL, and DL are anticipated to enhance predictive accuracy and contextual understanding as we transition into the next phase of intelligent finance. Integrating alternative data sources has expanded the scope of ML to include geolocation signals, social media sentiments, satellite imagery, and macroeconomic indicators. These data fusion models offer strategic advantages to financial analysts and traders by providing real-time insights into market dynamics and investor behavior. XAI and ethical AI will be essential in establishing trust and guaranteeing impartiality in automated financial judgments in the near future. Counterfactual reasoning, Shapley Additive exPlanation (SHAP) values, and causal inferences are essential for interpreting black-box ML models. In addition, the convergence of ML with blockchain, privacy-enhancing technologies, and secure multiparty computation is facilitating the development of privacy-preserving ML applications in the financial services sector (Chen, 2024). This safeguards sensitive data, while ensuring regulatory compliance. The operational paradigm of financial firms has been redefined by the transition to cloud-native, API-driven, and automated ML pipelines on the infrastructural front. However, several obstacles remain, including regulatory uncertainty, adversarial assaults, algorithmic opacity, and data bias. To fully realize the potential of ML in finance, future research must concentrate on trustworthy AI frameworks that balance performance and ethics, autonomous ML agents for real-time market adaptation, and cross-sector collaboration to standardize AI governance in financial ecosystems.

7.2 Future scope in QML

At the forefront of computational finance, QML provides novel paradigms to address intractable issues with improved speed, parallelism, and efficiency. The integration of quantum hardware with ML is expected to produce innovative tools for portfolio optimization, fraud detection, and risk analysis as quantum hardware continues to evolve. In portfolio performance metrics, QSVMs outperform their classical counterparts. QNNs with fewer trainable parameters match or exceed classical models in credit scoring, and hybrid architectures such as QCNNs achieving superior results in credit default prediction tasks through VQCs and entangled data encoding (Feretzakis et al., 2024). Several quantum-enhanced models have demonstrated this potential. Additionally, quantum transformers and QGNNs are emerging as the next-generation generative and graph-based models capable of revealing intricate financial dependencies.

7.2.1. Future trends and research directions

The subsequent prospective trends and research priorities are essential for transitioning from experimentation to real-world adoption:

Hybrid quantum–classical models: Hybrid architectures that capitalize on the strengths of both domains are generated through the integration of classical ML and quantum models. For example, classical DL can perform preprocessing, with the quantum layers being responsible for feature extraction or optimization. These frameworks are particularly effective in environments restricted by quantum-hardware limitations.

Federated and privacy-preserving QML: In quantum finance, data privacy must be addressed, particularly in multi-institutional collaborations. Federated QML can facilitate the decentralized training of models across financial entities without the exchange of sensitive data. This guarantees secure and compliant innovation by integrating it with quantum cryptographic techniques (e.g., quantum homomorphic encryption).

Quantum financial data encoding techniques: The encoding of classical financial data for quantum processing is a critical research bottleneck. To efficiently manage high-dimensional time-series financial data, researchers are investigating advancements in density matrix-based representations, Q-sample encoding, and

amplitude encoding. However, domain-specific encoding standards continue to pose challenges. Roadmap for industry adoption: The roadmap for industry adoption includes the following:

- Investigating real-time risk analytics and quantum-aware fraud detection by conducting pilot projects.
- Creating domain-specific QML libraries (e.g., by utilizing TensorFlow Quantum, PennyLane, or Qiskit Finance).
- Retraining financial analysts and quants in constructing a quantum-ready infrastructure.

Potential breakthroughs and open problems: QML, despite its immense potential, must overcome the following obstacles: (1) Noise and decoherence in NISQ devices; (2) Inadequate scalable quantum circuits for large-scale financial datasets, and (3) Evaluation and benchmarking standards. The following are the potential sources of future breakthroughs: quantum kernel methods that learn nonlinear structures from financial data, QAOA-based solvers for option pricing and risk management, and quantum Bayesian learning for adaptive financial inference. QML has the potential to redefine the architecture of financial intelligence fundamentally as the quantum ecosystem matures and interdisciplinary collaboration occurs.

8. Conclusion

The integration of QML into the financial realm extends beyond traditional schemes, marking a distinctive advancement beyond the potential of classical ML. Although Classical ML has transformed the systems of fraud detection, forecasting, and credit scoring, it is now increasingly constrained by limitations in computational scalability, interpretability, and real-time adaptability as markets become increasingly complex. QML overcomes these limitations by exploiting the quantum mechanical effects of superposition, entanglement, and quantum parallelism to provide either an exponential or polynomial speedup for computationally intensive algorithmic tasks. This performance boost is prominent in algorithmic trading within portfolio optimization, option pricing, HFT, and large-scale market simulation, where quantum algorithms such as QSVM, QNN, QAOA, and quantum-enhanced clustering allow high-dimensional modeling and complex pattern recognition that classical methods cannot achieve. A unique set of enhancements offered by QML includes quantum

kernel methods for enhanced anomaly detection, hybrid quantum–classical DL for credit risk analysis, and quantum RL for real-time strategy optimization in rapidly changing markets. Real-world advantages of implementing these advancements include reduced trade latency, enhanced fraud-detection accuracy, greater portfolio risk-return balance, and greater predictive capability during times of distress. In addition, QML is concerned with the next version of financial intelligence that it can enable in practice. Realizing this potential entails solving persistent challenges: encoding classical financial data into quantum states efficiently and without losing significance; scaling to massive datasets on an NISQ device; regulatory compliance of algorithmic decision-making; and improving the interpretability of quantum models to generate institutional trust. Therefore, future research should prioritize three directions.

- (1) Tailoring domain-specific QML algorithms to financial time series, market microstructures, and derivative pricing.
- (2) Providing robust hybrid quantum–classical architectures for near-term performance improvements while offsetting the limitations of quantum hardware.
- (3) Explainable quantum AI frameworks for transparent, fair, and ethical compliance in financial decision making

In the far future, while quantum hardware is still being developed, the ML industry and QML will merge to construct financial processes not only by building incrementally on one another but also by supervising the inception of a quantum-intelligent age that can model computational efficiency, with perfect predictive capability on market behavior and toughness in markets.

CRedit authorship contribution statement

M. Gunaleswaran: Conceptualization, Investigation, Methodology, Writing of the original draft. **K. Ramanan:** Conceptualization, Supervision. **H. Leelavathi:** Methodology, Writing, Review & Editing. **S. K. Muthusundar:** Formal analysis, Validation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that may appear to have influenced the work reported in this paper.

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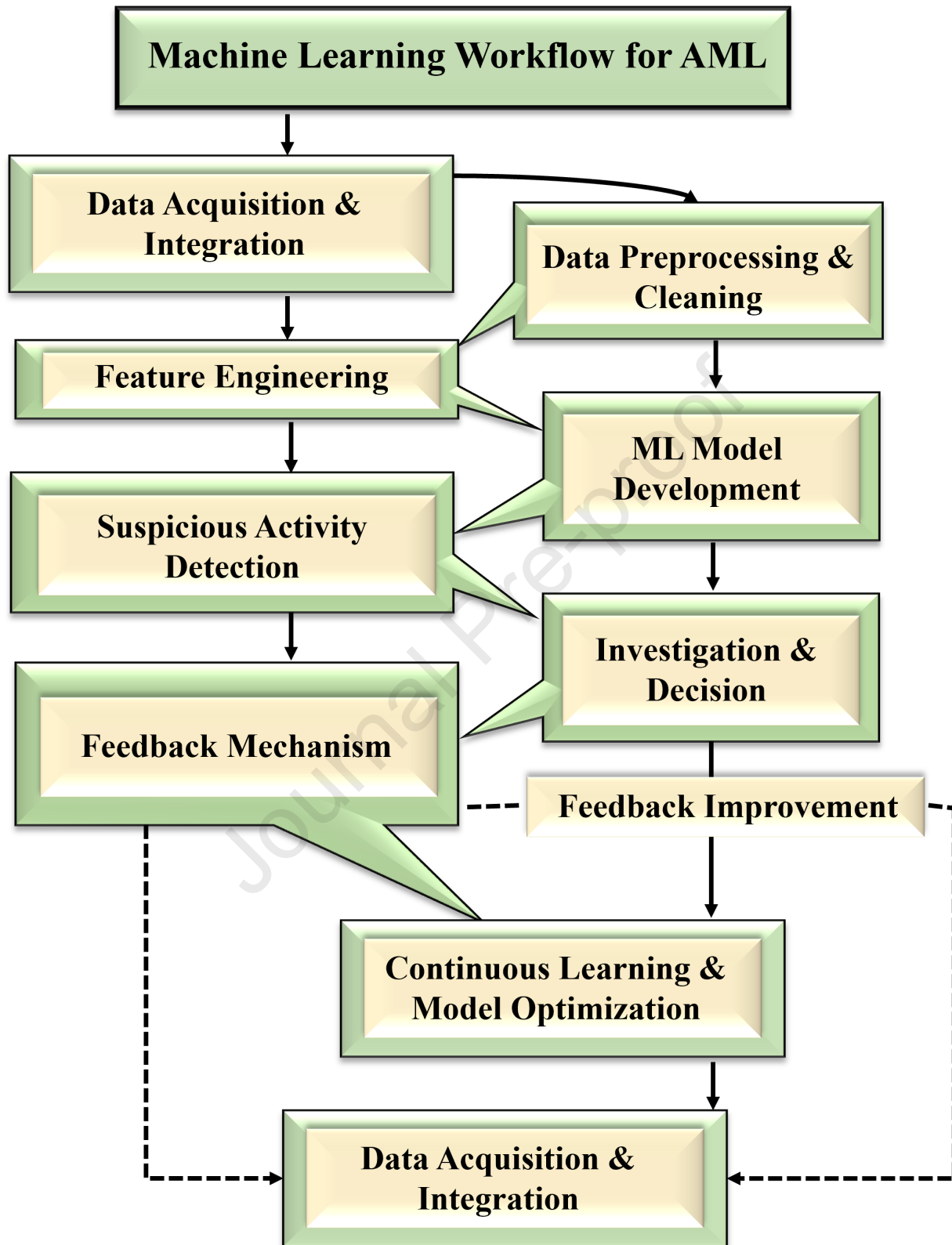
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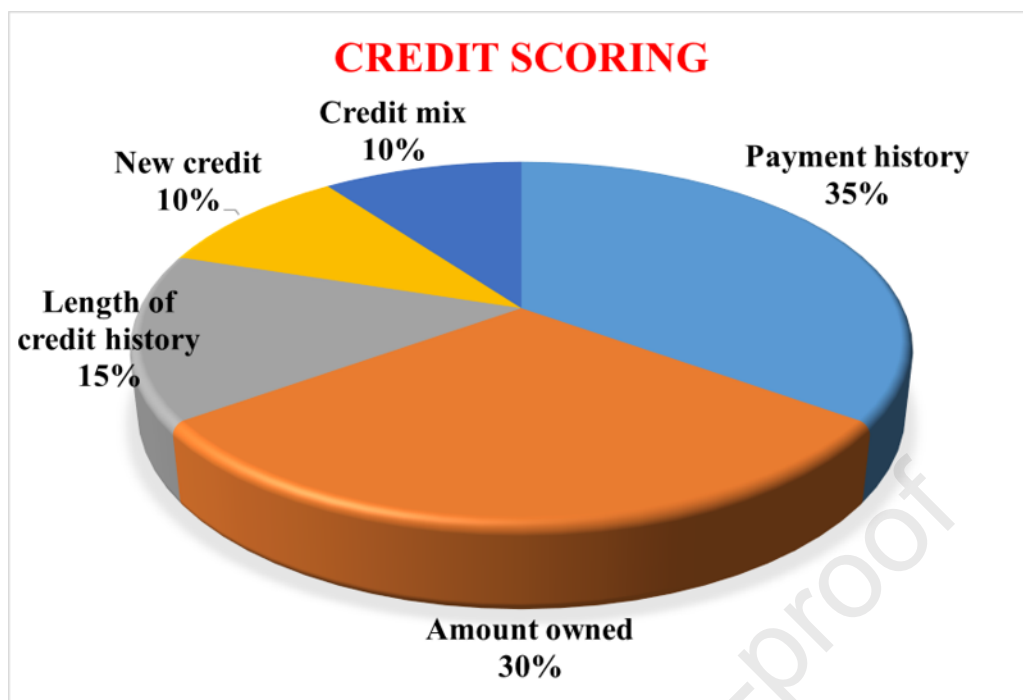
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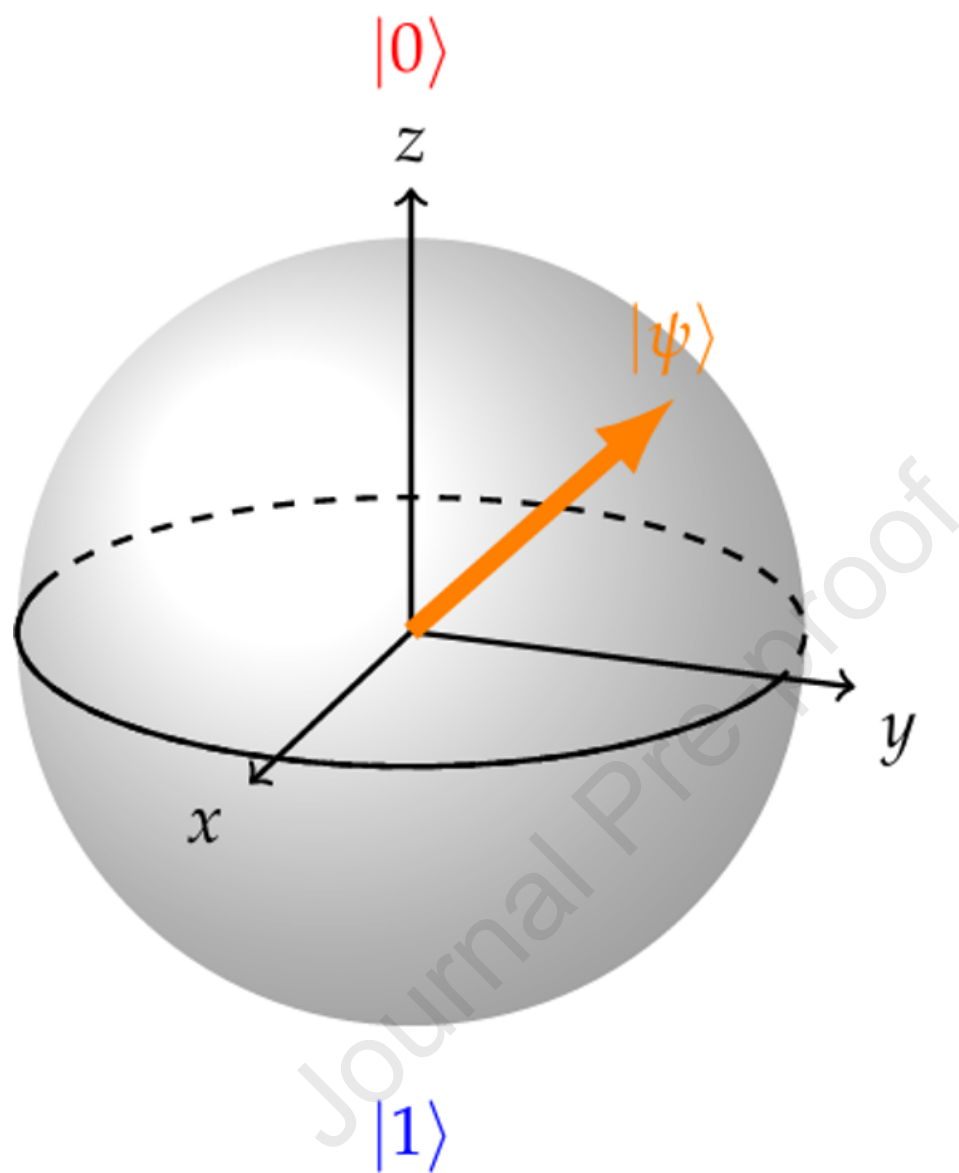
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- * Quantum Machine Learning (QML) leverages unique quantum properties like superposition, entanglement, and parallelism to overcome the computational limitations of traditional ML approaches.
- * Techniques such as Quantum Support Vector Machines (QSVM), Quantum Neural Networks (QNN), and quantum clustering demonstrate significant advancements in efficiency and accuracy for financial applications.
- .
- * QML showcases transformative potential in solving complex financial problems, including fraud detection, portfolio optimization, and market forecasting, with greater accuracy and computational efficiency.
- * The potential for hybrid quantum-classical models to stabilize hardware and advance quantum algorithms offers a pathway to overcome current limitations, paving the way for scalable and robust financial systems.
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Declaration of interests

☒ The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.